COMP3670/6670
Introduction to Machine Learning
Week 10a: Singular Value Decomposition & Low-Rank Approximation

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Goals of this lecture

- Describe a rectangular matrix using SVD.
- Find a best low-rank approximation and calculate its error.
- Interpret matrix factors as user and item representations.
- Fit a rating model without treating missing ratings as zero.
- Use validation to choose hyperparameters and assess predictions on held-out ratings.

SVD uses two sets of orthonormal directions

- The spectral theorem covers real symmetric square matrices.
- A user-item table can be rectangular: the numbers of users and items need not agree.
- For $\mathbf{A} \in \mathbb{R}^{m \times d}$, inputs lie in $\mathbb{R}^d$ and outputs lie in $\mathbb{R}^m$.
- SVD uses an orthonormal list in each space, rather than the same eigenbasis for both.

Singular value decomposition. Every nonzero matrix $\mathbf{A} \in \mathbb{R}^{m \times d}$ of rank r can be written as

$$\mathbf{A} \;=\; \mathbf{U}_r \boldsymbol{\Sigma}_r \mathbf{V}_r^{\mathsf{T}} \;=\; \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^{\mathsf{T}},$$

where

$$\mathbf{U}_r \in \mathbb{R}^{m \times r}, \qquad \boldsymbol{\Sigma}_r \in \mathbb{R}^{r \times r}, \qquad \mathbf{V}_r \in \mathbb{R}^{d \times r}.$$

The columns $\mathbf{u}_i$ of $\mathbf{U}_r$ and $\mathbf{v}_i$ of $\mathbf{V}_r$ form two orthonormal lists, and

$$\boldsymbol{\Sigma}_r \;=\; \operatorname{diag}(\sigma_1, \ldots, \sigma_r), \qquad \sigma_1 \geq \cdots \geq \sigma_r > 0.$$

The σ_i are the positive singular values. Retaining only these gives a compact SVD.

Write the column vectors explicitly:

$$\mathbf{U}_r = \begin{pmatrix} | & & | \\ \mathbf{u}_1 & \cdots & \mathbf{u}_r \\ | & & | \end{pmatrix}, \qquad \mathbf{V}_r = \begin{pmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_r \\ | & & | \end{pmatrix}.$$

The $\mathbf{u}_i$ are left singular vectors; the $\mathbf{v}_i$ are right singular vectors.

$$\mathbf{U}_r^\mathsf{T}\mathbf{U}_r = \mathbf{V}_r^\mathsf{T}\mathbf{V}_r = \mathbf{I}_r.$$

These matrices need not be square. In general, $\mathbf{U}_r\mathbf{U}_r^\mathsf{T}$ and $\mathbf{V}_r\mathbf{V}_r^\mathsf{T}$ are *not identity matrices*.

Example 1. Suppose $\mathbf{A}$ records users' ratings for items, with every rating known. Rows of $\mathbf{U}_r$ and $\mathbf{V}_r$ give user and item embeddings.

The indices $1, 2, \ldots, r$ label latent factors: patterns that may reflect aspects such as genre or style. Learned factors need not have a clear human interpretation.

$$\mathbf{A} = \begin{pmatrix} | & & | \\ \mathbf{u}_1 & \cdots & \mathbf{u}_r \\ | & & | \end{pmatrix} \begin{pmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r \end{pmatrix} \begin{pmatrix} | & & | \\ \mathbf{v}_1 & \cdots & \mathbf{v}_r \\ | & & | \end{pmatrix}^{\mathsf{T}}.$$

Example 1 (cont'd). Writing these rows as $\mathbf{p}_i^\mathsf{T}$ and $\mathbf{q}_j^\mathsf{T}$,

$$a_{ij} \;=\; \langle \mathbf{p}_i, \mathbf{\Sigma}_r \mathbf{q}_j \rangle \;=\; \sum_{\ell=1}^{r} \sigma_\ell p_{i\ell} q_{j\ell}.$$

This is a weighted dot product: the singular value σ_ℓ weights the contribution of coordinate ℓ.

Matching signs add to the score; opposite signs reduce it.

Recall that cosine similarity compares vector directions after normalizing their lengths. Here the rating score is weighted and unnormalized.

Example 1 (cont'd). The user and item embeddings predict a rating through the weighted dot product

$$a_{ij} \;=\; \sum_{\ell=1}^{r} \sigma_\ell p_{i\ell} q_{j\ell}.$$

Take

$$\mathbf{p}_i \;=\; \begin{pmatrix} 0.6 \\ 0.2 \end{pmatrix}, \quad \mathbf{q}_j \;=\; \begin{pmatrix} 0.5 \\ -0.4 \end{pmatrix}, \quad \mathbf{\Sigma}_2 \;=\; \begin{pmatrix} 10 & 0 \\ 0 & 2 \end{pmatrix}.$$

The predicted rating is

$$a_{ij} \;=\; 10(0.6)(0.5) + 2(0.2)(-0.4) \;=\; 3 - 0.16 \;=\; 2.84.$$

Proposition 1. Each singular-vector pair satisfies

$$\mathbf{A}\mathbf{v}_i \;=\; \sigma_i \mathbf{u}_i, \qquad \mathbf{A}^\mathsf{T}\mathbf{u}_i \;=\; \sigma_i \mathbf{v}_i.$$

Consequently,

$$\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{v}_i \;=\; \sigma_i^2 \mathbf{v}_i, \qquad \mathbf{A}\mathbf{A}^\mathsf{T}\mathbf{u}_i \;=\; \sigma_i^2 \mathbf{u}_i.$$

The positive eigenvalues of both matrices are $\sigma_1^2, \ldots, \sigma_r^2$, counted with multiplicity.

The remaining eigenvalues are zero: $d - r$ for $\mathbf{A}^\mathsf{T}\mathbf{A}$ and $m - r$ for $\mathbf{A}\mathbf{A}^\mathsf{T}$.

Proof outline. Expand the SVD and use orthonormality:

$$\mathbf{A}\mathbf{v}_i \;=\; \left(\sum_{j=1}^{r} \sigma_j \mathbf{u}_j \mathbf{v}_j^{\mathsf{T}}\right) \mathbf{v}_i \;=\; \sum_{j=1}^{r} \sigma_j \mathbf{u}_j \left\langle \mathbf{v}_j, \mathbf{v}_i \right\rangle \;=\; \sigma_i \mathbf{u}_i.$$

Orthonormality gives $\langle \mathbf{v}_j, \mathbf{v}_i \rangle = 0$ for $j \neq i$ and 1 for $j = i$, so only the $j = i$ term remains.

Similarly, $\mathbf{A}^{\mathsf{T}} \mathbf{u}_i = \sigma_i \mathbf{v}_i$.

Apply $\mathbf{A}^{\mathsf{T}}$ or $\mathbf{A}$ to obtain the eigenvector equations.

- $\mathbf{A}^\mathsf{T}\mathbf{A}$ is symmetric and positive semidefinite:

$$\mathbf{x}^\mathsf{T}\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x} \;=\; \|\mathbf{A}\mathbf{x}\|_2^2 \;\geq\; 0.$$

- Find its positive eigenvalues λ_i and corresponding orthonormal eigenvectors $\mathbf{v}_i$. Then

$$\sigma_i \;=\; \sqrt{\lambda_i}, \qquad \mathbf{u}_i \;=\; \frac{\mathbf{A}\mathbf{v}_i}{\sigma_i}.$$

- This explains SVD using the spectral theorem. Numerical software can compute SVD directly without forming $\mathbf{A}^\mathsf{T}\mathbf{A}$.

Example 2. Consider

$$\mathbf{A} = \begin{pmatrix} 20 & 10 \\ 8 & 19 \\ -2 & 14 \end{pmatrix}, \qquad \mathbf{A}^{\mathsf{T}}\mathbf{A} = \begin{pmatrix} 468 & 324 \\ 324 & 657 \end{pmatrix}.$$

The right-hand matrix has orthonormal eigenvectors

$$\mathbf{v}_1 = \frac{1}{5}\begin{pmatrix} 3 \\ 4 \end{pmatrix}, \qquad \mathbf{v}_2 = \frac{1}{5}\begin{pmatrix} 4 \\ -3 \end{pmatrix},$$

with eigenvalues 900 and 225, respectively.

Hence the singular values are $\sigma_1 = 30$ and $\sigma_2 = 15$.

Example 2 (cont'd).

$$\mathbf{A} = \begin{pmatrix} 20 & 10 \\ 8 & 19 \\ -2 & 14 \end{pmatrix}, \quad \mathbf{v}_1 = \frac{1}{5}\begin{pmatrix} 3 \\ 4 \end{pmatrix}, \quad \mathbf{v}_2 = \frac{1}{5}\begin{pmatrix} 4 \\ -3 \end{pmatrix}.$$

Using $\sigma_1 = 30$ and $\sigma_2 = 15$,

$$\mathbf{u}_1 = \frac{\mathbf{A}\mathbf{v}_1}{30} = \frac{1}{3}\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \qquad \mathbf{u}_2 = \frac{\mathbf{A}\mathbf{v}_2}{15} = \frac{1}{3}\begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}.$$

Their squared norms are both 1, and their dot product is $(4 - 2 - 2)/9 = 0$.

Example 2 (cont'd). Collecting the vectors and singular values gives

$$\underbrace{\begin{pmatrix} 20 & 10 \\ 8 & 19 \\ -2 & 14 \end{pmatrix}}_{\mathbf{A}} = \underbrace{\begin{bmatrix} \frac{1}{3} \begin{pmatrix} 2 & 2 \\ 2 & -1 \\ 1 & -2 \end{pmatrix} \end{bmatrix}}_{\mathbf{U}_2} \underbrace{\begin{pmatrix} 30 & 0 \\ 0 & 15 \end{pmatrix}}_{\mathbf{\Sigma}_2} \underbrace{\begin{bmatrix} \frac{1}{5} \begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix} \end{bmatrix}^{\mathsf{T}}}_{\mathbf{V}_2^{\mathsf{T}}}.$$

For any input vector, the calculation has three stages:

$$\mathbf{x} \in \mathbb{R}^2 \xrightarrow{\mathbf{V}_2^{\mathsf{T}}} \text{input coordinates} \xrightarrow{\mathbf{\Sigma}_2} \text{scaled coordinates} \xrightarrow{\mathbf{U}_2} \mathbf{A}\mathbf{x} \in \mathbb{R}^3.$$

Each unit direction $\mathbf{v}_i$ is mapped to $\sigma_i \mathbf{u}_i$, so σ_i measures its stretching factor.

Truncation gives a best low-rank approximation

Outer products and squared error. For nonzero column vectors $\mathbf{u} \in \mathbb{R}^m$ and $\mathbf{v} \in \mathbb{R}^d$,

$$\mathbf{u}\mathbf{v}^\mathsf{T} = (u_i v_j)_{i,j}$$

is an outer product. It has rank one: every column is a multiple of $\mathbf{u}$, and at least one is nonzero.

The Frobenius norm is

$$\|\mathbf{A}\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{d} a_{ij}^2}.$$

Thus $\|\mathbf{A} - \mathbf{B}\|_F^2$ is the total squared entrywise error.

Truncated SVD. The compact SVD separates a matrix into rank-one contributions:

$$\mathbf{A} \;=\; \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^{\mathsf{T}}.$$

Keep the first k terms, where $1 \le k \le r$:

$$\mathbf{A}_k \;=\; \sum_{i=1}^{k} \sigma_i \mathbf{u}_i \mathbf{v}_i^{\mathsf{T}} \;=\; \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^{\mathsf{T}}.$$

Every column of $\mathbf{A}_k$ lies in $\mathrm{Col}(\mathbf{U}_k)$. Conversely, $\mathbf{A}_k \mathbf{v}_i = \sigma_i \mathbf{u}_i$ for $i \le k$.

Eckart-Young theorem (1936). For every matrix $\mathbf{B}$ of rank at most k,

$$\|\mathbf{A} - \mathbf{B}\|_F^2 \geq \|\mathbf{A} - \mathbf{A}_k\|_F^2 = \sum_{i=k+1}^{r} \sigma_i^2.$$

The truncated SVD gives a best rank-at-most-k approximation under total squared error.

The error is the sum of the squares of the discarded singular values.

When $k = r$, the reconstruction is exact.

Example 3. Recall from Example 2:

$$\mathbf{A} = \begin{pmatrix} 20 & 10 \\ 8 & 19 \\ -2 & 14 \end{pmatrix}, \quad \mathbf{u}_1 = \frac{1}{3}\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{v}_1 = \frac{1}{5}\begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$

Its singular values are 30 and 15. Keeping only the first gives

$$\mathbf{A}_1 = 30\mathbf{u}_1\mathbf{v}_1^{\mathsf{T}} = \begin{pmatrix} 12 & 16 \\ 12 & 16 \\ 6 & 8 \end{pmatrix}.$$

Every column is a multiple of $(2, 2, 1)^{\mathsf{T}}$, so this approximation has rank one.

Example 3 (cont'd). The original matrix and rank-one approximation:

$$\mathbf{A} = \begin{pmatrix} 20 & 10 \\ 8 & 19 \\ -2 & 14 \end{pmatrix}, \qquad \mathbf{A}_1 = \begin{pmatrix} 12 & 16 \\ 12 & 16 \\ 6 & 8 \end{pmatrix}.$$

The residual is

$$\mathbf{A} - \mathbf{A}_1 = \begin{pmatrix} 8 & -6 \\ -4 & 3 \\ -8 & 6 \end{pmatrix}.$$

Its total squared error is $64 + 36 + 16 + 9 + 64 + 36 = 225 = 15^2$. The discarded singular value was 15. No rank-one matrix gives a smaller total squared error.

- Storing $\mathbf{U}_k$, the diagonal of $\mathbf{\Sigma}_k$ and $\mathbf{V}_k$ takes $k(m+d+1)$ numbers, rather than md.
- Storage decreases only when k is sufficiently small.
- Repeated singular values can allow different equally good truncated approximations.
- Mathematical optimality concerns squared reconstruction error; it does not guarantee success on an ML task.

Recommender systems use user and item factors

Latent-factor rating model. Let $\mathbf{R} \in \mathbb{R}^{n \times m}$ record ratings from n users for m items. Assign

$$\mathbf{p}_i \in \mathbb{R}^k \text{ to user } i, \qquad \mathbf{q}_j \in \mathbb{R}^k \text{ to item } j, \qquad \widehat{r}_{ij} \;=\; \langle \mathbf{p}_i, \mathbf{q}_j \rangle .$$

Stack these vectors as rows:

$$\mathbf{P} \in \mathbb{R}^{n \times k}, \qquad \mathbf{Q} \in \mathbb{R}^{m \times k}, \qquad \widehat{\mathbf{R}} \;=\; \mathbf{P}\mathbf{Q}^{\mathsf{T}}.$$

The prediction matrix has rank at most k.

For a complete table, $\mathbf{P} = \mathbf{U}_k \mathbf{\Sigma}_k$ and $\mathbf{Q} = \mathbf{V}_k$ give $\widehat{\mathbf{R}} = \mathbf{R}_k$. The singular values have been absorbed into the user factors.

Example 4. Six users give scores out of 10 to four films:

$$\mathbf{R} \;=\; \begin{pmatrix} 10 & 8 & 4 & 2 \\ 8 & 10 & 2 & 4 \\ 9 & 9 & 3 & 3 \\ 4 & 2 & 10 & 8 \\ 2 & 4 & 8 & 10 \\ 3 & 3 & 9 & 9 \end{pmatrix}.$$

Gaussian elimination gives rank 3. Its positive singular values are

$$\sigma_1 \;=\; 12\sqrt{6}, \qquad \sigma_2 \;=\; 6\sqrt{6}, \qquad \sigma_3 \;=\; 4.$$

The first three users prefer the first two films; the last three prefer the last two.

Example 4 (cont'd). One compact SVD is

$$\mathbf{R} = \underbrace{\begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{2} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & -\frac{1}{2} \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{1}{2} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{1}{2} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & 0 \end{pmatrix}}_{\mathbf{U}_3} \underbrace{\begin{pmatrix} 12\sqrt{6} & 0 & 0 \\ 0 & 6\sqrt{6} & 0 \\ 0 & 0 & 4 \end{pmatrix}}_{\mathbf{\Sigma}_3} \underbrace{\left[\frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix} \right]}_{\mathbf{V}_3^\mathsf{T}}.$$

Rows of $\mathbf{U}_3$ embed users; rows of $\mathbf{V}_3$ embed items. As in Example 1, each rating is their weighted dot product, with weights $12\sqrt{6}$, $6\sqrt{6}$ and 4.

Example 4 (cont'd). The singular values are $12\sqrt{6}$, $6\sqrt{6}$ and 4. Keep the first two, and absorb them into the user factors:

$$\mathbf{P} \;=\; \mathbf{U}_2\mathbf{\Sigma}_2 \;=\; \begin{pmatrix} 12 & 6 \\ 12 & 6 \\ 12 & 6 \\ 12 & -6 \\ 12 & -6 \\ 12 & -6 \end{pmatrix}, \qquad \mathbf{Q} \;=\; \mathbf{V}_2 \;=\; \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & -1 \\ 1 & -1 \end{pmatrix}.$$

For user 1, $\mathbf{p}_1 = (12, 6)^\mathsf{T}$. Films 1 and 3 have factors

$$\mathbf{q}_1 \;=\; \frac{1}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{q}_3 \;=\; \frac{1}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Their predicted scores are $12/2 + 6/2 = 9$ and $12/2 - 6/2 = 3$.

Example 4 (cont'd). The rank-two reconstruction is

$$\mathbf{R}_2 \;=\; \mathbf{P}\mathbf{Q}^{\mathsf{T}} \;=\; \begin{pmatrix} 9 & 9 & 3 & 3 \\ 9 & 9 & 3 & 3 \\ 9 & 9 & 3 & 3 \\ 3 & 3 & 9 & 9 \\ 3 & 3 & 9 & 9 \\ 3 & 3 & 9 & 9 \end{pmatrix}.$$

Its total squared error and root mean squared error (RMSE) are

$$\|\mathbf{R} - \mathbf{R}_2\|_F^2 \;=\; 4^2 \;=\; 16, \qquad \mathrm{RMSE} \;=\; \sqrt{\frac{16}{24}} \;\approx\; 0.82.$$

The main group preferences remain, but smaller within-group differences disappear.

Missing ratings require fitting only observed entries

- Most users rate only a small fraction of the available items.
- An unrecorded rating is unknown, not zero.
- Ordinary SVD needs a numerical value in every position. Filling blanks with zero invents ratings.
- Keep the latent-factor model, but fit it using only observed training pairs.

Observed-rating objective. Let Ω be the set of observed training pairs. Fit the factors by minimizing

$$L(\mathbf{P}, \mathbf{Q}) = \sum_{(i,j)\in\Omega} \left(r_{ij} - \langle \mathbf{p}_i, \mathbf{q}_j \rangle\right)^2 + \lambda \left(\sum_{i=1}^{n} \|\mathbf{p}_i\|_2^2 + \sum_{j=1}^{m} \|\mathbf{q}_j\|_2^2 \right), \qquad \lambda \geq 0.$$

Only training ratings appear in the error sum.

The squared-norm term is a regularization penalty. As in SVM, fitting the data competes with keeping parameter norms small.

- The factor count k and penalty strength λ are hyperparameters, selected using validation ratings.
- The learned $\mathbf{P}$ and $\mathbf{Q}$ need not have orthonormal columns.
- For a fully observed table and $\lambda = 0$, truncated SVD gives a global minimizer when $k \leq \operatorname{rank}(\mathbf{R})$.
- With missing entries or a positive penalty, that conclusion no longer follows from Eckart-Young.
- Regularization can discourage overfitting, but does not guarantee accurate predictions.

Alternating least squares. Fix the item factors. Each user's predicted ratings are then linear in that user's factor vector.

For a chosen k and $\lambda > 0$:

1. Initialize the factors with small random values, not all zeros.
2. Fix $\mathbf{Q}$ and fit each $\mathbf{p}_i$ using that user's training ratings and squared-norm penalty.
3. Fix $\mathbf{P}$ and fit each $\mathbf{q}_j$ using that item's training ratings and squared-norm penalty.
4. Repeat until the objective changes little or an iteration limit is reached.

- Exact minimization in each alternating step cannot increase the training objective.
- The joint problem is generally not convex. Different initializations can give different solutions; global optimality is not guaranteed.
- A user or item with no training ratings has no data from which the model can learn its factor vector.
- This is a cold-start problem. Our examples evaluate users and items that have training ratings.

Example 5. Recall from Example 4: the 6×4 complete rating table had rank 3, and its rank-two reconstruction had RMSE about 0.82.

Now suppose only these 18 ratings are observed:

$$\begin{pmatrix} 10 & 8 & ? & 2 \\ 8 & ? & 2 & 4 \\ 9 & 9 & 3 & ? \\ ? & 2 & 10 & 8 \\ 2 & 4 & ? & 10 \\ 3 & ? & 9 & 9 \end{pmatrix}.$$

Each question mark is unknown. The original hidden values from Example 4 must *not be used for fitting or selection*.

Example 5 (cont'd). There are 18 observed ratings. Hold out two for validation:

$$r_{12} = 8, \qquad r_{43} = 10.$$

Fit on the other 16 ratings and compute

$$\mathrm{RMSE}_{\mathrm{val}} = \sqrt{\frac{(8-\widehat{r}_{12})^2 + (10-\widehat{r}_{43})^2}{2}}.$$

Try $k \in \{1,2,3\}$ and $\lambda \in \{0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5\}$.

Each fit uses ten random initializations, retaining the smallest training objective. Choose (k, λ) by validation RMSE, not training error.

Example 5 (cont'd). Fitting used 16 ratings and validation used the two ratings 8 and 10. Validation selects $k = 2$, $\lambda = 0.01$, with RMSE about 1.22.

Refit on all 18 observed ratings to predict the six missing entries:

(i, j)	$(1,3)$	$(2,2)$	$(3,4)$	$(4,1)$	$(5,3)$	$(6,2)$
predicted	-1.50	7.33	5.61	-0.41	11.00	4.79
original	4	10	3	4	8	3

The original values are shown only for comparison. They were excluded from fitting and selection.

These predictions are unconstrained dot-product scores, so they can lie

Predicting missing MovieLens ratings

MovieLens 100K. A small movie-rating dataset from GroupLens:

- $100{,}000$ ratings, 943 users and $1{,}682$ films.
- Integer ratings from 1 to 5; only about 6.3% of user-film pairs are observed.
- The accompanying `movielens.py` downloads and caches the data.
- Store observed triples (i, j, r_{ij}), not zeros in the unknown positions.

The script fits the observed-rating objective, not an SVD of a zero-filled matrix.

Dataset: `https://grouplens.org/datasets/movielens/100k/`.

Harper and Konstan (2015), *The MovieLens Datasets: History and Context*.

Training, validation and testing.

1. Split into $80{,}000$ training, $10{,}000$ validation and $10{,}000$ test ratings, retaining training ratings for every user and film.
2. Try $k \in \{5, 10, 15, 20, 25, 30\}$ and $\lambda \in \{0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1, 2, 5\}$: 54 candidates.
3. Each fit uses one reproducible random initialization and 15 alternating sweeps. Select by validation RMSE.
4. Refit on the $90{,}000$ training and validation ratings; evaluate on test ratings excluded from fitting and selection.

The mean-rating baseline predicts the mean of those $90{,}000$ ratings for every user-film pair. Test ratings do not enter that mean.

Clipping and test RMSE. Ratings lie in $[1, 5]$, so clip the raw dot-product predictions:

$$\widehat{r}_{ij}^{\,\mathrm{clip}} \;=\; \min\{5, \max\{1, \langle \mathbf{p}_i, \mathbf{q}_j \rangle\}\}.$$

For held-out test pairs $\mathcal{T}$,

$$\mathrm{RMSE}_{\mathrm{test}} \;=\; \sqrt{\frac{1}{|\mathcal{T}|} \sum_{(i,j)\in\mathcal{T}} \left(r_{ij} - \widehat{r}_{ij}^{\,\mathrm{clip}}\right)^2}.$$

Apply clipping to both validation and test predictions, without changing the fitting objective.

MovieLens results. 54 candidates were compared by validation, followed by refitting on $90{,}000$ ratings. The selected pair is $k = 5$, $\lambda = 5$.

quantity	RMSE
validation, clipped	0.9153
test, clipped	0.9305
same model's test, unclipped	0.9321
mean-rating baseline, test	1.1320

Fan et al. (ICLR 2024) report 0.8767 using *AEMC-NE*, an autoencoder-based method. Their result averages 20 random $90/10$ splits: context, not a controlled comparison.

From predicted ratings to recommendations. For user 1, the script ranks films with no recorded rating by their raw scores. Displayed scores are clipped:

film	predicted rating
A Close Shave	5.00
Dr. Strangelove	5.00
Paths of Glory	4.95
Ran	4.88
Casablanca	4.88

Their true ratings are unknown, so we cannot calculate prediction errors for these pairs.

Takeaways

- SVD describes a rectangular matrix using left and right orthonormal directions.
- Truncation minimizes total squared reconstruction error; the error is $\sum_{i>k} \sigma_i^2$.
- Low-dimensional user and item factors predict ratings through dot products.
- Missing ratings are unknown, not zero: fit only observed training entries. Validation chooses hyperparameters. Held-out test ratings evaluate prediction, not guaranteed recommendation quality.